

# Sri Lanka Institute of Information Technology



IT2130 – Operating Systems and System Administration  
Year 2, Semester 2- 2026

Practical Answer Submission

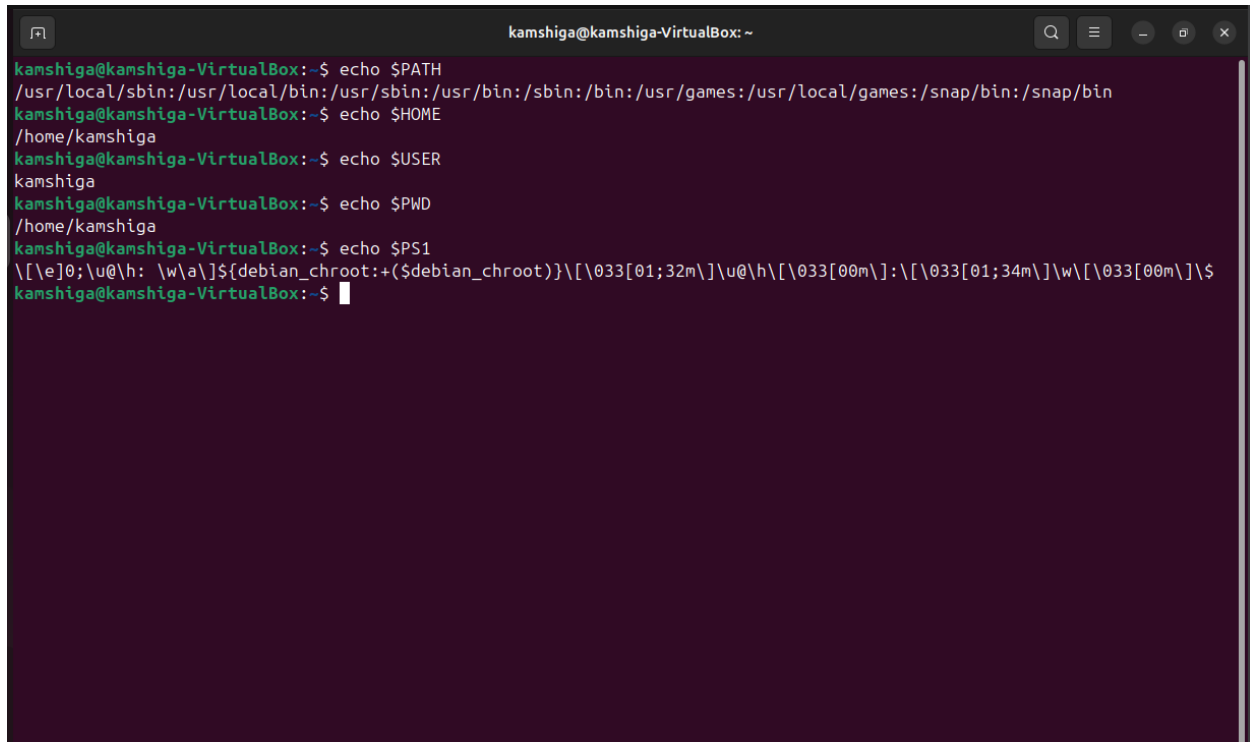
Practical Sheet No: 10

|                     |                      |
|---------------------|----------------------|
| Student ID          | IT24101365           |
| Student Name        | Kamshiga Ganesan     |
| Campus/ Center name | Northern UNI         |
| Specialization      | Software Engineering |
| Batch No            | 1                    |

## Activity 1

Write down the output for following commands: (Remember Unix is case sensitive, therefore type them as they are.)

echo \$PATH , echo \$HOME , echo \$USER, echo \$PWD, echo \$PS1

A terminal window titled 'kamshiga@kamshiga-VirtualBox: ~' with a dark purple background. It shows the execution of five 'echo' commands to display system environment variables. The output for \$PATH is a long string of directory paths. \$HOME is /home/kamshiga. \$USER is kamshiga. \$PWD is /home/kamshiga. \$PS1 is a complex escape sequence for the shell prompt.

```
kamshiga@kamshiga-VirtualBox:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
kamshiga@kamshiga-VirtualBox:~$ echo $HOME
/home/kamshiga
kamshiga@kamshiga-VirtualBox:~$ echo $USER
kamshiga
kamshiga@kamshiga-VirtualBox:~$ echo $PWD
/home/kamshiga
kamshiga@kamshiga-VirtualBox:~$ echo $PS1
\[e]0;\u@h: \w\a\]${debian_chroot:+($debian_chroot)}\[033[01;32m]\u@h\[033[00m\]:\[033[01;34m]\w\[033[00m\]\$
kamshiga@kamshiga-VirtualBox:~$
```

a) Did you assign the values that you noted down for the above variables?

- No

b) If not, who assigned those values for the above variables?

- The operating system and the shell automatically assigned those values when the user logged in.

## Activity 2

Assign and print following variables

```
kamshiga@kamshiga-VirtualBox: ~  
kamshiga@kamshiga-VirtualBox:~$ labclass="This is OSSA Lab Class"  
kamshiga@kamshiga-VirtualBox:~$ echo $labclass  
This is OSSA Lab Class  
kamshiga@kamshiga-VirtualBox:~$ num=`expr 13 + 5`  
kamshiga@kamshiga-VirtualBox:~$ echo $num  
18  
kamshiga@kamshiga-VirtualBox:~$ num=`expr 13 \* 5`  
kamshiga@kamshiga-VirtualBox:~$ echo $num  
65  
kamshiga@kamshiga-VirtualBox:~$ num1=`expr $num \/ 7`  
kamshiga@kamshiga-VirtualBox:~$ echo $num1  
9  
kamshiga@kamshiga-VirtualBox:~$ num1=`expr $num \% 7`  
kamshiga@kamshiga-VirtualBox:~$ echo $num1  
2  
kamshiga@kamshiga-VirtualBox:~$ num1=`expr $num1 + 8`  
kamshiga@kamshiga-VirtualBox:~$ echo $num1  
10  
kamshiga@kamshiga-VirtualBox:~$
```

```
kamshiga@kamshiga-VirtualBox: ~  
kamshiga@kamshiga-VirtualBox:~$ nano usrinfo  
kamshiga@kamshiga-VirtualBox:~$ chmod 777 usrinfo  
kamshiga@kamshiga-VirtualBox:~$ ls -l usrinfo  
-rwxrwxrwx 1 kamshiga kamshiga 173 Apr  9 16:20 usrinfo  
kamshiga@kamshiga-VirtualBox:~$ chmod 770 usrinfo  
kamshiga@kamshiga-VirtualBox:~$ ls -l usrinfo  
-rwxrwx--- 1 kamshiga kamshiga 173 Apr  9 16:20 usrinfo  
kamshiga@kamshiga-VirtualBox:~$
```

```
GNU nano 7.2                               usrinfo
#!/bin/bash

name="Kamshiga Ganesan"
address="No 522,Kasthuriyaar Road, Jaffna"
country="Sri Lanka"

echo "Name   : $name"
echo "Address: $address"
echo "Country: $country"
```

Write down the commands that can be used to execute a shell script,

a) in which user doesn't have execution permission

b) in which user has execution permission

```
kamshiga@kamshiga-VirtualBox: ~ × kamshiga@kamshiga-VirtualBox: ~ × kamshiga@kamshiga-VirtualBox: ~ × kamshiga@kamshiga-VirtualBox: ~ ×
kamshiga@kamshiga-VirtualBox:~$ nano usrinfo
kamshiga@kamshiga-VirtualBox:~$ chmod 777 usrinfo
kamshiga@kamshiga-VirtualBox:~$ ls -l usrinfo
-rwxrwxrwx 1 kamshiga kamshiga 173 Apr  9 16:20 usrinfo
kamshiga@kamshiga-VirtualBox:~$ chmod 770 usrinfo
kamshiga@kamshiga-VirtualBox:~$ ls -l usrinfo
-rwxrwx--- 1 kamshiga kamshiga 173 Apr  9 16:20 usrinfo
kamshiga@kamshiga-VirtualBox:~$ sh usrinfo
Name   : Kamshiga Ganesan
Address: No 522,Kasthuriyaar Road, Jaffna
Country: Sri Lanka
kamshiga@kamshiga-VirtualBox:~$ ./usrinfo
Name   : Kamshiga Ganesan
Address: No 522,Kasthuriyaar Road, Jaffna
Country: Sri Lanka
kamshiga@kamshiga-VirtualBox:~$ █
```

### Activity 3

Make a directory scripts inside your working directory. Change your directory to new directory (by typing cd scripts).

```
kamshiga@kamshiga-Vi... x kamshiga@kamshiga-Vi... x kamshiga@kamshiga-Vi... x kamshiga@kamshiga-Vi... x kamshiga@kamshiga-Vi... x
kamshiga@kamshiga-VirtualBox:~$ mkdir scripts
kamshiga@kamshiga-VirtualBox:~$ cd scripts
kamshiga@kamshiga-VirtualBox:~/scripts$ nano script1
kamshiga@kamshiga-VirtualBox:~/scripts$ sh script1
Welcome to SLIIT Server
Today is:Thu Apr 9 04:57:13 PM +0530 2026
kamshiga@kamshiga-VirtualBox:~/scripts$
```

```
GNU nano 7.2 script1
#!/bin/bash
echo "Welcome to SLIIT Server"
echo "Today is:$(date)"

```

Why can't you type script1 to run it? (Hint: Type `ls -l script1` and check the file permissions.)

Run `chmod u+x script1` to enable the rights for

```
kamshiga@kamshiga-Vi... x kamshiga@kamshiga-Vi... x kamshiga@kamshiga-Vi... x kamshiga@kamshiga-Vi... x kamshiga@kamshiga-Vi... x
kamshiga@kamshiga-VirtualBox:~$ mkdir scripts
kamshiga@kamshiga-VirtualBox:~$ cd scripts
kamshiga@kamshiga-VirtualBox:~/scripts$ nano script1
kamshiga@kamshiga-VirtualBox:~/scripts$ sh script1
Welcome to SLIIT Server
Today is:Thu Apr  9 04:57:13 PM +0530 2026
kamshiga@kamshiga-VirtualBox:~/scripts$ ls -l script1
-rw-rw-r-- 1 kamshiga kamshiga 69 Apr  9 16:56 script1
kamshiga@kamshiga-VirtualBox:~/scripts$ chmod u+x script1
kamshiga@kamshiga-VirtualBox:~/scripts$ ls -l script1
-rwxrw-r-- 1 kamshiga kamshiga 69 Apr  9 16:56 script1
kamshiga@kamshiga-VirtualBox:~/scripts$ ./script1
Welcome to SLIIT Server
Today is:Thu Apr  9 05:00:46 PM +0530 2026
kamshiga@kamshiga-VirtualBox:~/scripts$
```